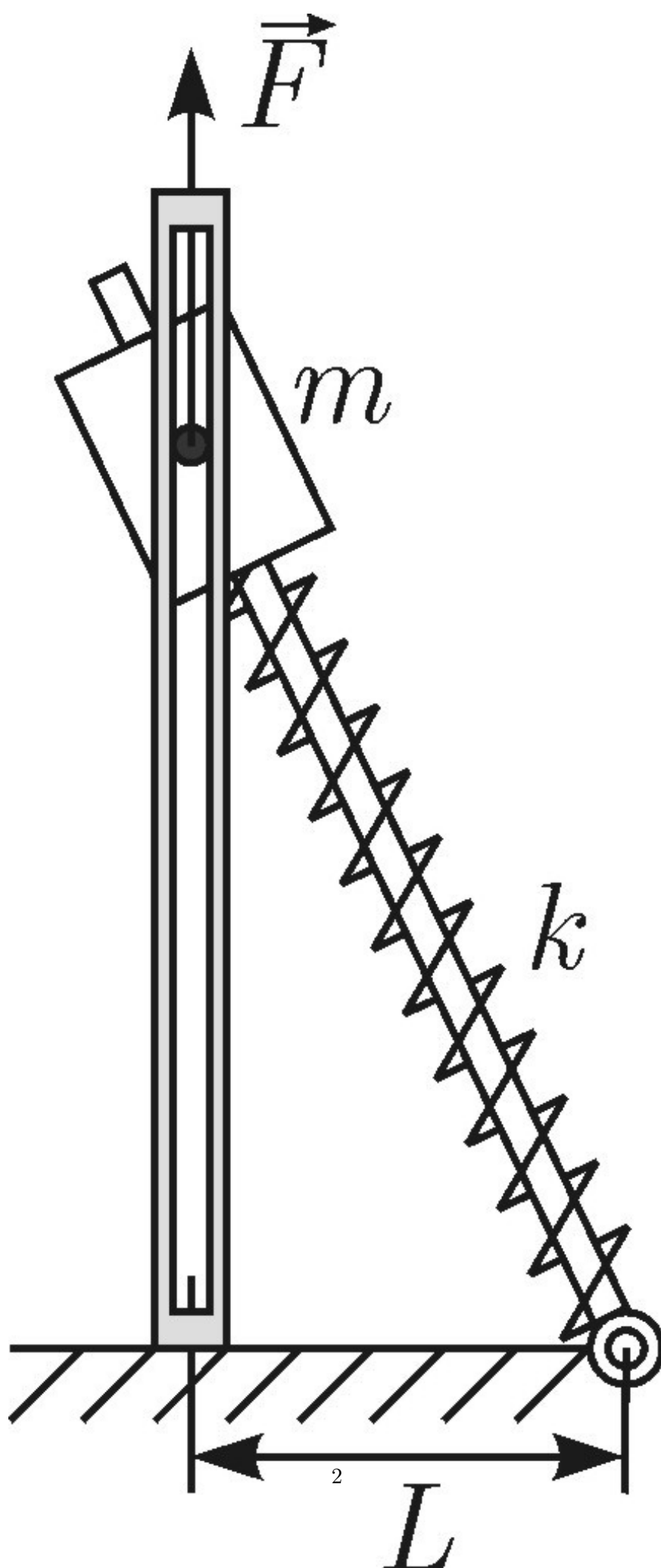


Detail

Y14 (2014) — English translation

A part with mass m can move without friction along vertical guides and rotate freely around a horizontal axis perpendicular to the guides and the drawing plane. The part has a through hole into which a smooth light rod is inserted, one end of which is hinged on a horizontal plane at a distance L from the base of the vertical guides (Fig. 1). Between the part and the hinge, a spring attached to them with a stiffness coefficient k is put on the rod. Free spring length $3L$. A light thread is tied to the part, which in the initial state is tensioned by a vertically upward force $F = mg$.

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- | | | |
|-----------|---|----------|
| A1 | What should be the minimum mass $m_{\min}$ of the part so that with a slow (quasi-static) decrease in force F , the part can move the rod to a horizontal position under the influence of gravity. Ignore friction. | 0 |
|-----------|---|----------|

If the force F is reduced to zero instantly, then the part can move the rod to a horizontal position with a smaller mass m_1 . The graph shows the dependence of the projection of the elastic force on the vertical axis, in relative units, on the distance from the load to the horizontal plane.

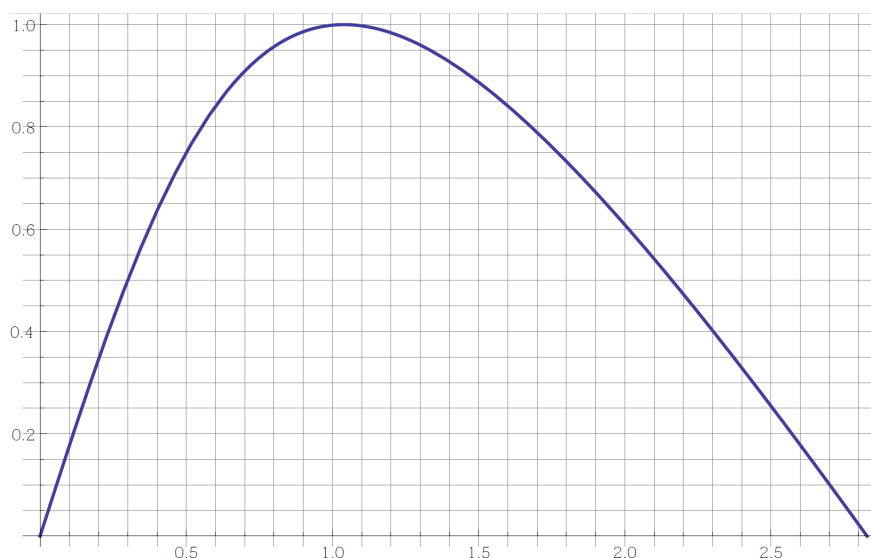


Figure 2: Figure 2.

- | | | |
|-----------|---|----------|
| A2 | Determine the mass of m_1 using the graph provided. | 0 |
|-----------|---|----------|

- | | | |
|-----------|---|----------|
| A3 | The force F was eliminated (the thread was cut), and the mass load m_1 was transferred to a stable equilibrium position. Determine the period of small oscillations near this position. | 0 |
|-----------|---|----------|

Source: <https://pho.rs/p/3963>